Are Fandango's Movie Ratings Inflated?

A plain-language walkthrough of a small data-analysis project

The Question

In 2015, journalist Walt Hickey noticed something odd: movies with genuinely bad reviews were still showing 4 or 5 stars on Fandango. He suspected the ratings were being inflated. This project re-runs his investigation using the same public data, to check whether the evidence really supports that claim — and whether anything changed after his article came out.

Why This Project Exists

It's a small project on purpose. The goal isn't to build something complicated — it's to practice the core skill any data or analyst role asks for: take a specific claim, pull real data, test it honestly, and explain what you found in plain English. That loop (question → data → test → conclusion) is the same one used in actual analyst work, just at a much smaller scale.

The Data

Two real datasets, both published by FiveThirtyEight (the data-journalism outlet Hickey worked for):

- **fandango_score_comparison.csv** — 146 films from 2015, each with Fandango's rating alongside Rotten Tomatoes, Metacritic, and IMDB, all converted onto the same 0–5 star scale so they're fairly comparable.
- **fandango_scrape.csv** — 510 films, showing the star rating Fandango *displays* on the page next to the *true* rating value hidden in the page's underlying HTML code.

A third dataset was added later for the before/after comparison: 214 popular films from 2016–2017, rated on the same four platforms, collected independently by a different analyst after Hickey's article was published.

How the Project Works, Step by Step

1. Load and inspect the data

The CSV files are loaded with pandas, and the first few rows are checked to understand what each column actually means — especially since different sites use different scales (0–100 for Rotten Tomatoes, 0–10 for IMDB, 0–5 for Fandango).

2. Explore with SQL

The data is loaded into a small SQLite database (a lightweight, file-free SQL database) and queried directly with SQL — simple AVG() and COUNT() queries to answer questions like "what's the average displayed rating vs. the true rating?" This is the same kind of query an analyst would write against a real company database.

3. Compare displayed stars to the true rating

For each film, the star rating shown to users is compared against the true rating hidden in the HTML. If Fandango were fair, these two numbers should match closely.

4. Compare Fandango to other platforms

Fandango's ratings are plotted against Rotten Tomatoes, Metacritic, and IMDB on the same scale, to see whether Fandango is unusually generous compared to everyone else.

5. Check before vs. after the article

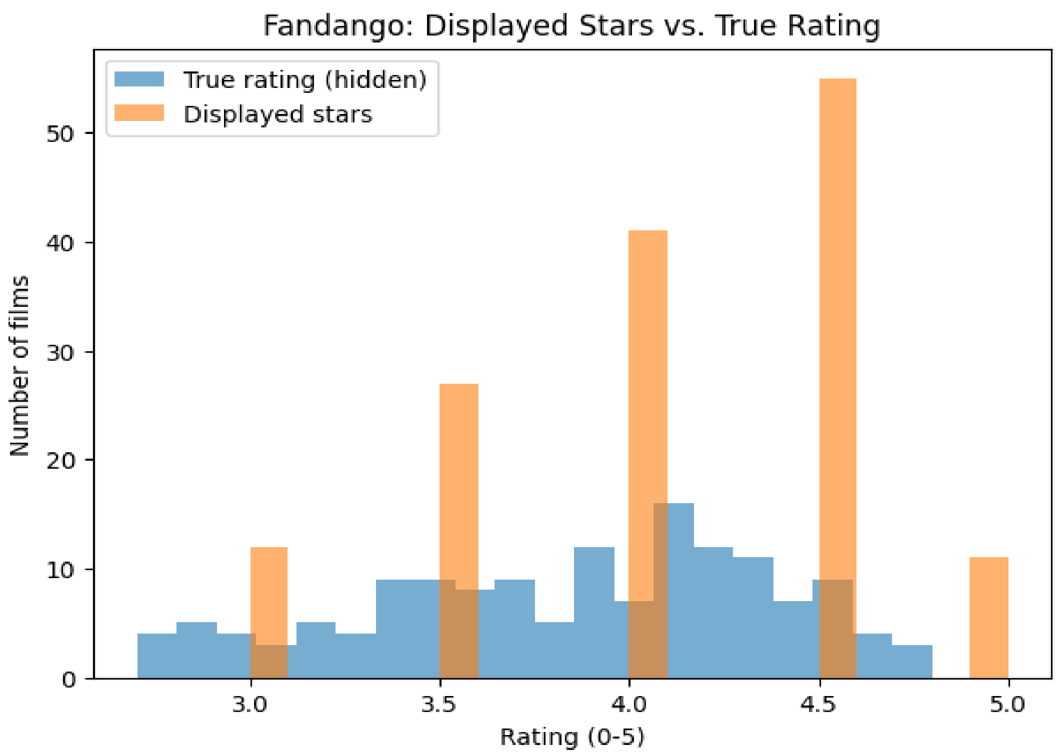
A second, independent dataset from 2016–2017 is brought in to see whether Fandango's average rating moved at all after the bias was publicly reported.

6. Write the conclusion

The last step is just plain-English: what did the numbers actually show, and does the data support the original claim?

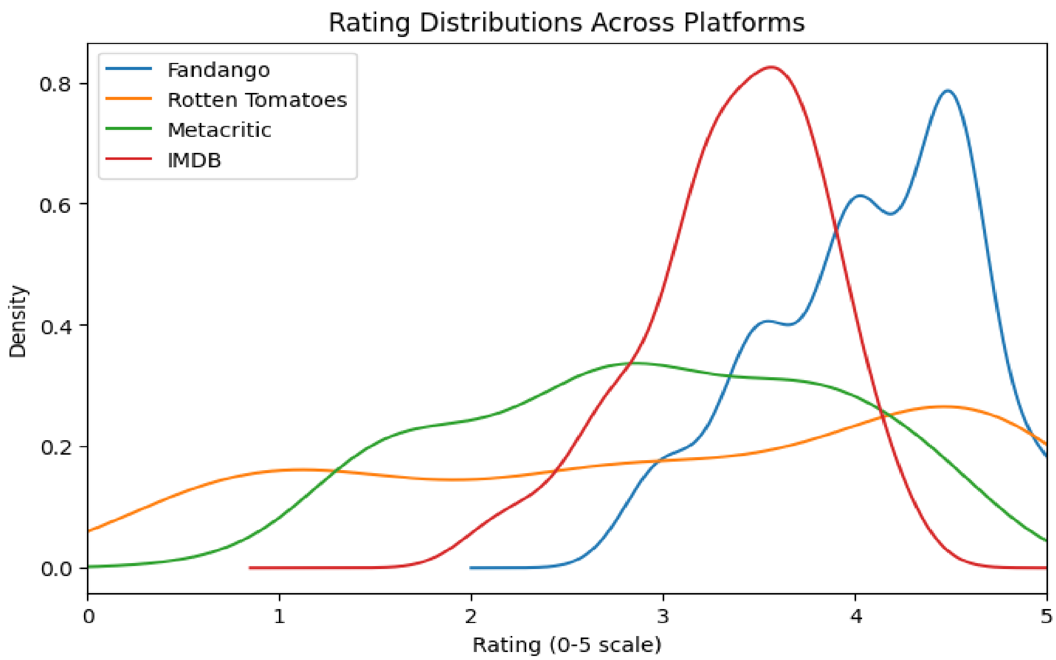
What the Data Actually Showed

Displayed rating vs. true rating (2015): the average displayed star rating was 4.09, compared to an average true rating of 3.85. In 130 of 146 films, the displayed rating was rounded up from the true value.



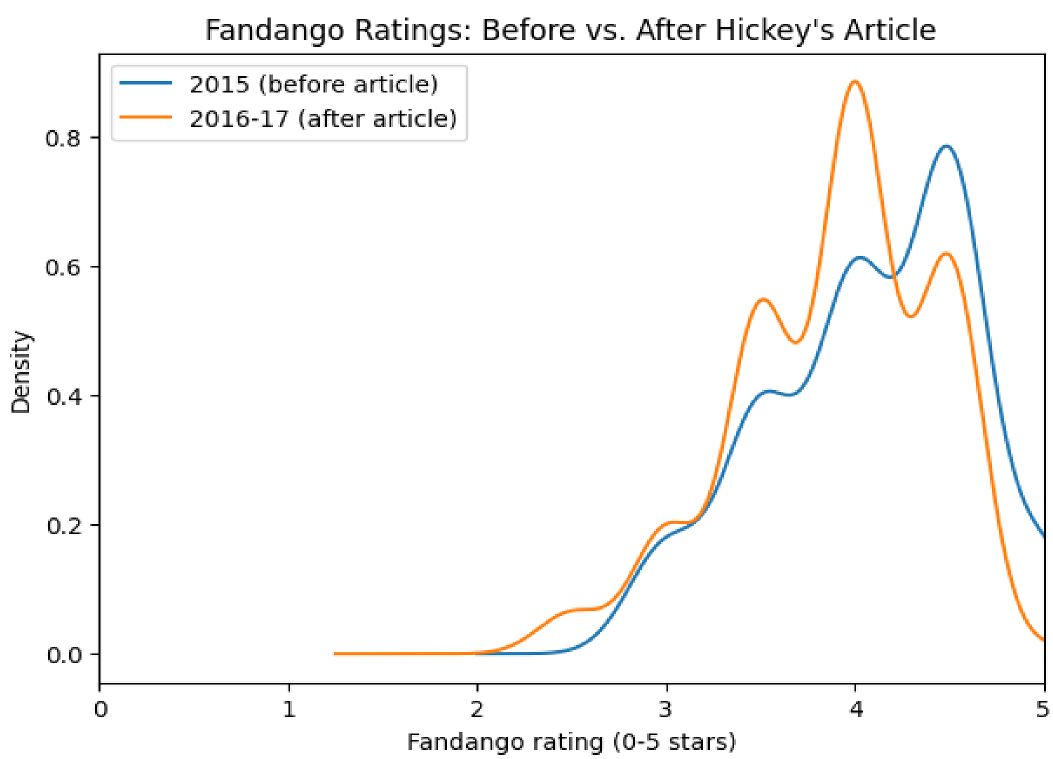
The orange bars (displayed stars) are shifted right compared to the blue bars (true rating) — a visual sign that ratings cluster higher once they're rounded for display.

Fandango vs. other platforms: Fandango's average (4.09) was clearly higher than Rotten Tomatoes (3.04), Metacritic (2.94), and IMDB (3.37) — all on the same 5-star scale.



Fandango's curve (blue) sits noticeably to the right of the other three platforms, meaning its ratings are concentrated much closer to 5 stars.

Before vs. after the article: using the independent 2016–17 dataset, the average Fandango rating only moved from 3.85 to 3.89 — a tiny change, suggesting the underlying behavior didn't shift much even after the bias was made public.



The 2015 and 2016–17 distributions (blue vs. orange) look very similar in shape and position, which is itself the finding: little visible change.

A Limitation Worth Knowing

The before/after comparison uses two datasets collected by different people, in different years, without perfectly identical methodology. It's the closest honest comparison the available public data allows — but it's not a perfectly controlled experiment, and that's worth saying out loud rather than glossing over.

Bottom Line

The data supports Hickey's original claim: Fandango's displayed ratings are consistently higher than both the true hidden rating and the ratings on every other platform checked. That pattern barely changed even after the bias was publicly reported.

Full code: fandango_analysis.ipynb | Data: fandango_score_comparison.csv, fandango_scrape.csv, movie_ratings_16_17.csv